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Use Case 1 - Sourcing Sustainable Materials for a Data Centre:

Use Case Title		Sourcing Sustainable Materials for a Data Centre
Use Case ID		UC-DATA-01
Goal in Context		User is looking for sustainable value from choosing components for a data centre.
Preconditions		User is signed up, authenticated and familiar with software, etc.
Success End Condition		User can make an educated and informed decision of what components to choose.
Failed End Condition		User is unable to make an insightful choice of component
Primary Actors		Chief Information Officer (CIO)
Secondary Actors		Database, Customer Support
Trigger		User logs in
Description	Step	Action
	1	User inputs their requirements to be read by the AI system.
	2	The AI system produces results and weights in factors including environmental, technological, economic and social factors.
	3	The user evaluates the options
Extensions	Step	Branching Action
	1a	The user may input a separate use case
	3a	The user may choose to contact customer support
	3b	The user may choose to view previous analysis
Sub-variations		Branching Action
	2	Responsive design may vary depending on screen resolution

Use Case 2 - Telecommunications Energy Management:

Use Case Title		Telecommunications Energy Management
Use Case ID		UC-TELE-02
Goal in Context		User is looking to reduce energy consumption to contribute to sustainability goals and lower cost
Preconditions		User is signed up, authenticated and familiar with software, etc. User has their infrastructural components known
Success End Condition		User can see their environmental impact from current infrastructure as well as recommendations on alternative solutions
Failed End Condition		User is unable to see their environmental impact or alternative solutions
Primary Actors		Energy Conservation Team of the Telecommunications Company
Secondary Actors		Database, Customer Support
Trigger		User logs in
Description	Step	Action
	1	User inputs their current components to be read by the AI system.
	2	The AI system produces results and weights factors including environmental, technological, economic and social factors.
	3	The user evaluates the results and alternative options
Extensions	Step	Branching Action
	1a	The user may input a separate use case
	3a	The user may choose to contact customer support
	3b	The user may choose to view previous analysis
Sub-variations		Branching Action
	2	Responsive design may vary depending on screen resolution

Use Case 3 - Data-driven sustainability in encouraging solar panel adoption

Background - Energy companies are incentivised to promote adoption of renewable energy sources like solar panels, in residential settings. This approach can benefit the environment, reduce carbon emissions, and help customers save money on their energy bills.

STEPS:

1) Data collection and analysis:

- Energy companies collect and analyse historical data on residential energy consumption patterns which includes data on when and how much energy is used in different households.
- Satellite imagery and geographical data can be used to assess the solar potential of individual households, considering facts like roof orientation, shading and geographical location.
- Customer profiles are generated to understand their preferences and financial situations.

2) Customer engagement:

- Data is used to offer personalised recommendations to homeowners about the benefits of installing solar panels - potential cost saving and environmental benefits.
- Data-driven energy saving tips are provided to homeowners, further reducing their energy consumption.

3) Monitor and feedback:

- Install systems to collect real-time data on energy production and consumption for customers with solar panels. Provide customers with easy-to-understand dashboards to track their energy savings and environmental impact.
- Use data analytics to provide homeowners with ways to optimise their energy usage and improve reliance on solar energy.

4) Performance based incentives:

- Offer performance-based incentives to energy companies based on the percentage of customers they successfully convert to solar energy.

5) Data sharing and research:

- Collaborate with research institutions to collect and share data on the environmental and economic impact of solar adoption, further building the case for sustainability.

Use Case 4 - Data-driven sustainability in encouraging firms to go paperless

Background - Incentivising firms to go completely paperless by designing a sustainability certification program. This program would assess, and reward businesses based on their paperless initiatives, making use of data-driven metrics to measure and track their progress.

STEPS:

1) Data collection and monitoring:

- Program will commence by collecting data from firms to understand their current paper usage.
- This data would include metrics such as volume of paper used, the number of printed documents, and the environmental impact associated with paper consumption.

2) Benchmarking and goal setting:

- Once the initial data is collected, the program would set benchmarks and goals for paper reduction.
- These goals would be tailored to the size and industry of the firm, ensuring that they are both challenging and achievable.
- Data - driven analysis would be used to determine the most effective targets for each business.

3) Feedback and guidance:

- Firms would receive ongoing feedback and guidance based on their data. The program would use data analytics to identify areas of improvement.

4) Rewards and certification:

- Firms that meet and exceed their paper reduction goals would receive sustainability certification. This certification could come with various benefits, such as marketing opportunities and public recognition.

5) Continuous improvement:

- Program would encourage participating firms to continue their paperless journey by setting increasingly ambitious goals over time. Data-driven analysis would help identify new opportunities for reducing paper consumption and improving sustainability.

Use Case 5 - Best practices for data-driven sustainability (general example)

Background - Company XYZ is a global technology firm committed to reducing its environmental footprint. They want to implement data-driven solutions to align their operations with sustainable development goals and reduce their environmental impact.

STEPS

1) Data collection and analysis:

- Company XYZ begins by collecting comprehensive data on its operations, energy consumption, waste generation, and supply chain activities.
- They utilise advanced sensors, IoT devices, and data analytics tools to monitor and analyse real-time data.

2) Sustainability benchmarking:

- Sustainability performance is benchmarked against industry standards and best practices are identified - targets are set after their data is compared to industry leaders.

3) Predictive analytics:

- Company XYZ implements predictive analytics models to forecast future environment impacts based on various scenarios and variables, such as market conditions, regulatory changes, and technological advancements.
- These models lead to better decision-making, minimisation of risks, and optimization of resource allocation.

4) Decision support systems:

- A support system is developed which provides real-time insights to employees, empowering them to make sustainable decisions in their daily operations.
- Employees can access data on energy usage, waste management, and resource consumption to drive informed decisions that reduce environmental impact.

5) Supply chain optimization:

- Company XYZ optimises its supply chain to reduce emissions, minimise waste, and source materials sustainably.
- They collaborate with suppliers who share similar goals.

6) Renewable energy integration:

- Organisation invests in renewable energy sources like solar and wind power to focus on reducing carbon emissions.

7) Continuous monitoring and reporting:

- Company XYZ continuously monitors its sustainability efforts, regularly collecting and analysing data to track progress toward sustainability goals.
- They generate sustainability reports, disclosing their environmental impact and progress to stakeholders.